

LoanWizard: Prototype Architecture (as built)

The system that runs in this repository today. Boxes map to real directories; dashed boxes are mocked or optional by design.

MVP · SHIPPED

BROWSER PERCEPTION

packages/perception · Stream A

Camera & mic capture
getUserMedia · 10s rolling audio buffer

Speech-to-text
Web Speech API → Whisper fallback

Liveness (TF.js)
BlazeFace · landmarks · blink · head-pose

Document OCR
Tesseract.js · Aadhaar & PAN, on-device

Age estimate
@vladmandic/face-api · vendored weights

Device fingerprint
fraud signal input

PerceptionEvents ▾ typed, dispatched to the session store

WEB APPLICATION

apps/web · Stream C · Next.js 14 App Router

Session state machine
start → interview → decision → accept

Operator Console UI
live gauges · co-pilot · offer card

Prisma audit writes
transcripts · CV signals · consent

Admin dashboards
sessions · fairness · drift · replay

i18n · SSE
English / Hindi · live updates

▾ POST /offer · server-computed from persisted session data ▲ Offer

ML DECISION SERVICE

apps/ml-service · Stream B · FastAPI

Risk MLP
Keras v1.2.0 · permutation importance

Fraud MLP
Keras v0.1.0

Persona classifier
rules v1.0.0

Policy engine
hard eligibility rules

Reason narrator
weighted codes + plain language

Drift · fairness · replay
replay on exact archived version

Bureau merge
mock CIBIL / Experian (adapter)

LLM narration
template default · Gemini optional

Whisper STT model
opt-in deploy toggle

▾ decision row + frozen input snapshot + drift sample

DATA PLANE

PostgreSQL · one database, two ORMs

Sessions & audit
Prisma (web)

Decisions & snapshots
SQLAlchemy (ML) · frozen for replay

Model version stamps
per-decision lineage

Real & running in this repo Mocked / optional by design (adapter or deploy toggle in place)

See docs/architecture-target.pdf for the target production platform.

LoanWizard · AI video loan origination · prototype architecture. Request path is server-authoritative: the client never submits the offer inputs directly, the ML service recomputes from persisted session data.

SHARED CONTRACTS

packages/contracts
Frozen TypeScript types, Zod schemas and mock data. Every stream consumes the same session, perception and decision shapes, so the three streams integrate without drift.

BUILD MODEL

Three parallel streams (A/B/C) against the frozen contract package, each developed independently and integrated through typed contracts.

DEPLOY TARGET

Two Docker Spaces (web + ML) on Hugging Face, Neon Postgres for the shared audit trail.

EXPLAINABILITY

Every offer ships reason codes + narrative; the input snapshot is frozen so the decision replays on the same model version.